

Numerical Methods

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Reference

J. Mathews and R. L. Walker, *Mathematical Methods of Physics*

I. INTERPOLATION

Suppose we have a two-column data set as given in Table I with equally spaced x values, where $x_i = x_0 + ih$.

A. Forward Differences

We define the difference operator Δ such that,

$$\Delta y_i = y_{i+1} - y_i \Rightarrow y_{i+1} = (1 + \Delta)y_i \Rightarrow y_i = (1 + \Delta)^i y_0, \quad (1)$$

which we can expand using,

$$\begin{aligned} \Delta^2 y_i &= \Delta(\Delta y_i) = \Delta y_{i+1} - \Delta y_i, \dots, \\ \Delta^n y_i &= \Delta(\Delta^{n-1} y_i) = \Delta^{n-1} y_{i+1} - \Delta^{n-1} y_i. \end{aligned} \quad (2)$$

To interpolate the value of y at some intermediate value x , such that $x_0 < x < x_n$, we use the same formula as the one used for points where data is available. I.e.,

$$y(x) = (1 + \Delta)^\alpha y_0, \quad (3)$$

where, $x = x_0 + \alpha h \Rightarrow \alpha = \frac{x - x_0}{h}$. Expanding, we get,

$$y(x) = y_0 + \alpha \Delta y_0 + \frac{\alpha(\alpha - 1)}{2!} \Delta^2 y_0 + \dots + \frac{\alpha(\alpha - 1) \dots (\alpha - n + 1)}{n!} \Delta^n y_0. \quad (4)$$

Note that even for non-integer α the series terminates since differences of the order greater than n are undefined for a data set with n entries. The above relation also simplifies to,

$$y(x) = y_0 + \frac{(x - x_0)}{h} \Delta y_0 + \frac{(x - x_0)(x - x_1)}{2!h^2} \Delta^2 y_0 + \dots + \frac{(x - x_0)(x - x_1) \dots (x - x_{n-1})}{n!h^n} \Delta^n y_0. \quad (5)$$

This is the Newton-Gregory forward difference interpolation formula.

When using this formula using pen and paper, it is useful to construct a table similar to the one shown below (5 data points are used for illustration).

B. Backward Differences

In the same way as above, we may define the backward difference operator ∇ such that,

$$\nabla y_i = y_i - y_{i-1} \Rightarrow y_{i-1} = (1 - \nabla)y_i \Rightarrow y_{n-i} = (1 - \nabla)^i y_n, \quad (6)$$

x_0	x_1	x_2	$\dots$	x_{n-2}	x_{n-1}	x_n
y_0	y_1	y_2	$\dots$	y_{n-2}	y_{n-1}	y_n

TABLE I: Two column data set.

x_0	y_0				
x_1	y_1	Δy_0			
x_2	y_2	Δy_1	$\Delta^2 y_0$	$\Delta^3 y_0$	
x_3	y_3	Δy_2	$\Delta^2 y_1$	$\Delta^3 y_1$	$\Delta^4 y_0$
x_4	y_4	Δy_3	$\Delta^2 y_2$		

TABLE II: Sample forward difference table for a data set with five entries.

which we can expand using,

$$\begin{aligned}\nabla^2 y_i &= \nabla(\nabla y_i) = \nabla y_i - \nabla y_{i-1}, \dots, \\ \nabla^n y_i &= \nabla(\nabla^{n-1} y_i) = \nabla^{n-1} y_i - \nabla^{n-1} y_{i-1}.\end{aligned}\quad (7)$$

To interpolate the value of y at some intermediate value x , such that $x_0 < x < x_n$, we use the same formula as the one used for points where data is available. We write $x = x_n + \alpha h \Rightarrow \alpha = \frac{x - x_n}{h}$. Thus,

$$y(x) = (1 - \nabla)^{-\alpha} y_n. \quad (8)$$

Expanding, we get,

$$y(x) = y_n + \alpha \nabla y_n + \frac{\alpha(\alpha+1)}{2!} \nabla^2 y_n + \dots + \frac{\alpha(\alpha+1) \dots (\alpha+n-1)}{n!} \nabla^n y_0. \quad (9)$$

As before, the above relation also simplifies to,

$$y(x) = y_n + \frac{(x - x_n)}{h} \nabla y_n + \frac{(x - x_n)(x - x_{n-1})}{2!h^2} \nabla^2 y_n + \dots + \frac{(x - x_n)(x - x_{n-1}) \dots (x - x_1)}{n!h^n} \nabla^n y_n. \quad (10)$$

This is the Newton-Gregory backward difference interpolation formula.

x_0	y_0				
x_1	y_1	∇y_1			
x_2	y_2	∇y_2	$\nabla^2 y_3$	$\nabla^3 y_3$	
x_3	y_3	∇y_3	$\nabla^2 y_3$	$\nabla^3 y_4$	$\nabla^4 y_4$
x_4	y_4	∇y_4	$\nabla^2 y_4$		

TABLE III: Sample backward difference table for a data set with five entries.

II. NUMERICAL DIFFERENTIATION

The forward, backward and central approximations for the numerical derivative of $f(x)$ at $x = x_0$ are given by,

$$f'(x_0) \approx \frac{f(x_0 + \delta x) - f(x_0)}{\delta x} \text{ (forward),} \quad (11)$$

$$f'(x_0) \approx \frac{f(x_0) - f(x_0 - \delta x)}{\delta x} \text{ (backward), and,} \quad (12)$$

$$f'(x_0) \approx \frac{f(x_0 + \delta x) - f(x_0 - \delta x)}{2\delta x} \text{ (central).} \quad (13)$$

The errors in the forward and backward formulae are of the order of δx (i.e., the ratio of the error to δx is neither negligible nor large compared to one). The error in the central difference formula is of the order of $(\delta x)^2$. Check these by doing Taylor expansions of $f(x_0 \pm \delta x)$ in the above expressions.

The numerical derivative may be calculated using the interpolation formulae. Differentiating Eq. 5, we get,

$$y'(x) = \frac{\Delta y_0}{h} + \left((x - x_0) + (x - x_1) \right) \frac{\Delta^2 y_0}{2!h^2} + \dots + (x - x_0)(x - x_1) \dots (x - x_{n-1}) \left(\sum_{i=0}^{n-1} \frac{1}{x - x_i} \right) \frac{\Delta^n y_0}{n!h^n}, \quad (14)$$

$$y'(x_0) = \frac{1}{h} \left(\Delta y_0 - \frac{\Delta^2 y_0}{2} + \dots + (-1)^{n-1} \frac{\Delta^n y_0}{n} \right). \quad (15)$$

This formula may be applied to any of the tabulated x -values to give,

$$y'(x_i) = \frac{1}{h} \left(\Delta y_i - \frac{\Delta^2 y_i}{2} + \cdots + (-1)^{n-i-1} \frac{\Delta^{n-i} y_i}{n-i} \right). \quad (16)$$

Similarly, using the backward interpolation formula, Eq. 10, we have,

$$y'(x_n) = \frac{1}{h} \left(\nabla y_n + \frac{\nabla^2 y_n}{2} + \cdots + \frac{\nabla^n y_n}{n} \right), \quad (17)$$

$$\Rightarrow y'(x_i) = \frac{1}{h} \left(\nabla y_i + \frac{\nabla^2 y_i}{2} + \cdots + \frac{\nabla^i y_i}{i} \right). \quad (18)$$

III. NUMERICAL INTEGRATION

Suppose we are given a data set as in Table I with equally spaced x -values, $x_i = x_0 + ih$. We are interested in the integral $\int_a^b y(x)dx$, where $a = x_0$ and $b = x_n$.

A. Trapezoidal Rule

The segment between x_i and x_{i+1} is assumed to be a straight line. The integral (area under the curve $y(x)$) is approximated by the sum of the areas of the trapeziums thereby formed. Thus,

$$\int_a^b y(x)dx \approx \sum_{i=0}^{n-1} \frac{1}{2} (y_i + y_{i+1})h = \frac{h}{2} (y_0 + (y_1 + y_2 + \cdots + y_{n-1}) + y_n). \quad (19)$$

B. Simpson's $\frac{1}{3}$ -Rule

In this approach, an even number of intervals is needed, i.e., an odd number of (x, y) -values. Consider two nearby strips with x -values x_i , $x_{i+1} = x_i + h$ and $x_{i+2} = x_i + 2h$.

$$\int_a^b y(x)dx = \sum_{\substack{i=0 \\ i \text{ even}}}^{n-2} \int_{x_i}^{x_{i+2}} y(x)dx \quad (20)$$

The unique quadratic which passes through these three points is assumed to be an approximation for the function in the region $[x_i, x_{i+2}]$. Let us take *Lagrange's formula* for the quadratic function.

$$\begin{aligned} y(x) &\approx y_i \frac{(x - x_{i+1})(x - x_{i+2})}{(x_i - x_{i+1})(x_i - x_{i+2})} + y_{i+1} \frac{(x - x_{i+2})(x - x_i)}{(x_{i+1} - x_{i+2})(x_{i+1} - x_i)} + y_{i+2} \frac{(x - x_i)(x - x_{i+1})}{(x_{i+2} - x_i)(x_{i+2} - x_{i+1})} \\ &= \frac{1}{h^2} \left(y_i \frac{(x - x_{i+1})(x - x_{i+2})}{2} - y_{i+1} (x - x_{i+2})(x - x_i) + y_{i+2} \frac{(x - x_i)(x - x_{i+1})}{2} \right) \end{aligned} \quad (21)$$

(Equivalently, Newton's forward/backward formula could be used. Try to do this derivation using one of them. It should be easy.)

Using integration by parts,

$$\begin{aligned} \int (x-a)(x-b)dx &= (x-a) \frac{(x-b)^2}{2} - \int \frac{(x-b)^2}{2} + \text{const}, \\ &= \frac{(x-a)(x-b)^2}{2} - \frac{(x-b)^3}{6} + \text{const}. \end{aligned} \quad (22)$$

$$\begin{aligned} \Rightarrow \int_{x_i}^{x_{i+2}} y(x)dx &\approx \frac{1}{h^2} \left[\frac{y_i}{2} \left(\frac{(x - x_{i+1})(x - x_{i+2})^2}{2} - \frac{(x - x_{i+2})^3}{6} \right) \right]_{x_i}^{x_{i+2}} - y_{i+1} \left(\frac{(x - x_{i+2})(x - x_i)^2}{2} - \frac{(x - x_i)^3}{6} \right) \right]_{x_i}^{x_{i+2}} \\ &\quad + \frac{y_{i+2}}{2} \left(\frac{(x - x_i)^2(x - x_{i+1})}{2} - \frac{(x - x_i)^3}{6} \right) \right]_{x_i}^{x_{i+2}} \end{aligned} \quad (23)$$

$$\begin{aligned}
\Rightarrow \int_{x_i}^{x_{i+2}} y(x) dx &\approx h \left[\frac{y_i}{2} \left(\frac{4}{2} + \frac{-8}{6} \right) - y_{i+1} \left(-\frac{8}{6} \right) + \frac{y_{i+2}}{2} \left(\frac{4}{2} - \frac{8}{6} \right) \right] \\
&\approx \frac{h}{3} (y_i + 4y_{i+1} + y_{i+2})
\end{aligned} \tag{24}$$

$$\begin{aligned}
\Rightarrow \int_a^b y(x) dx &= \sum_{\substack{i=0 \\ i \text{ even}}}^{n-2} \int_{x_i}^{x_{i+2}} y(x) dx \approx \sum_{\substack{i=0 \\ i \text{ even}}}^{n-2} \frac{h}{3} (y_i + 4y_{i+1} + y_{i+2}) \\
\Rightarrow \int_a^b y(x) dx &\approx \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})].
\end{aligned} \tag{25}$$

IV. NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS

Here, we will discuss the simplest method, i.e., Euler's method. We will only consider initial value problems. A first-order ordinary differential equation (ODE) is given by,

$$\dot{x} = f(x, t), \quad x(0) = x_0. \tag{26}$$

Choosing a time step, δt , we can write,

$$x(t + \Delta t) = x(t) + f(x(t), t) \delta t. \tag{27}$$

Thus, knowing the initial value of x , the value at all times can be obtained. The above method also applies to a set of first order ODEs, written in vector form.

$$\dot{\vec{x}} = \vec{f}(\vec{x}, t), \quad \vec{x}(0) = \vec{x}_0, \tag{28}$$

$$\Rightarrow \vec{x}(t + \Delta t) = \vec{x}(t) + \vec{f}(\vec{x}(t), t) \delta t. \tag{29}$$

An n^{th} order ODE is given by,

$$\frac{d^n x}{dt^n} = f\left(x, \dot{x}, \ddot{x}, \dots, \frac{d^{n-1}x}{dt^{n-1}}\right), \tag{30}$$

$$x(0) = c_0, \quad \dot{x}(0) = c_1, \dots, \left. \frac{d^{n-1}x}{dt^{n-1}} \right|_{t=0} = c_{n-1} \tag{31}$$

This may be written as n first order ODEs. We create an n -component vector whose first component is x and the remaining components are its first $(n - 1)$ derivatives. Thus, we get the following n first order ODEs.

$$\begin{aligned}
\dot{x}_0 &= x_1, \\
\dot{x}_1 &= x_2, \\
&\vdots \\
\dot{x}_{n-2} &= x_{n-1}, \\
\dot{x}_{n-1} &= f(x_0, x_1, \dots, x_{n-1})
\end{aligned} \tag{32}$$

For example, Newton's second law for a particle experiencing a position, velocity and time-dependent force and moving along one dimension can be expressed either as,

$$\ddot{x} = f(x, \dot{x}, t), \quad x(0) = 0, \quad \dot{x}(0) = 0, \tag{33}$$

or in the form of first order ODEs as,

$$\begin{aligned}
\dot{x} &= v, \\
\dot{v} &= f(x, v, t), \quad x(0) = 0, \quad v(0) = 0.
\end{aligned} \tag{34}$$

This form may be solved using Eq. 29.